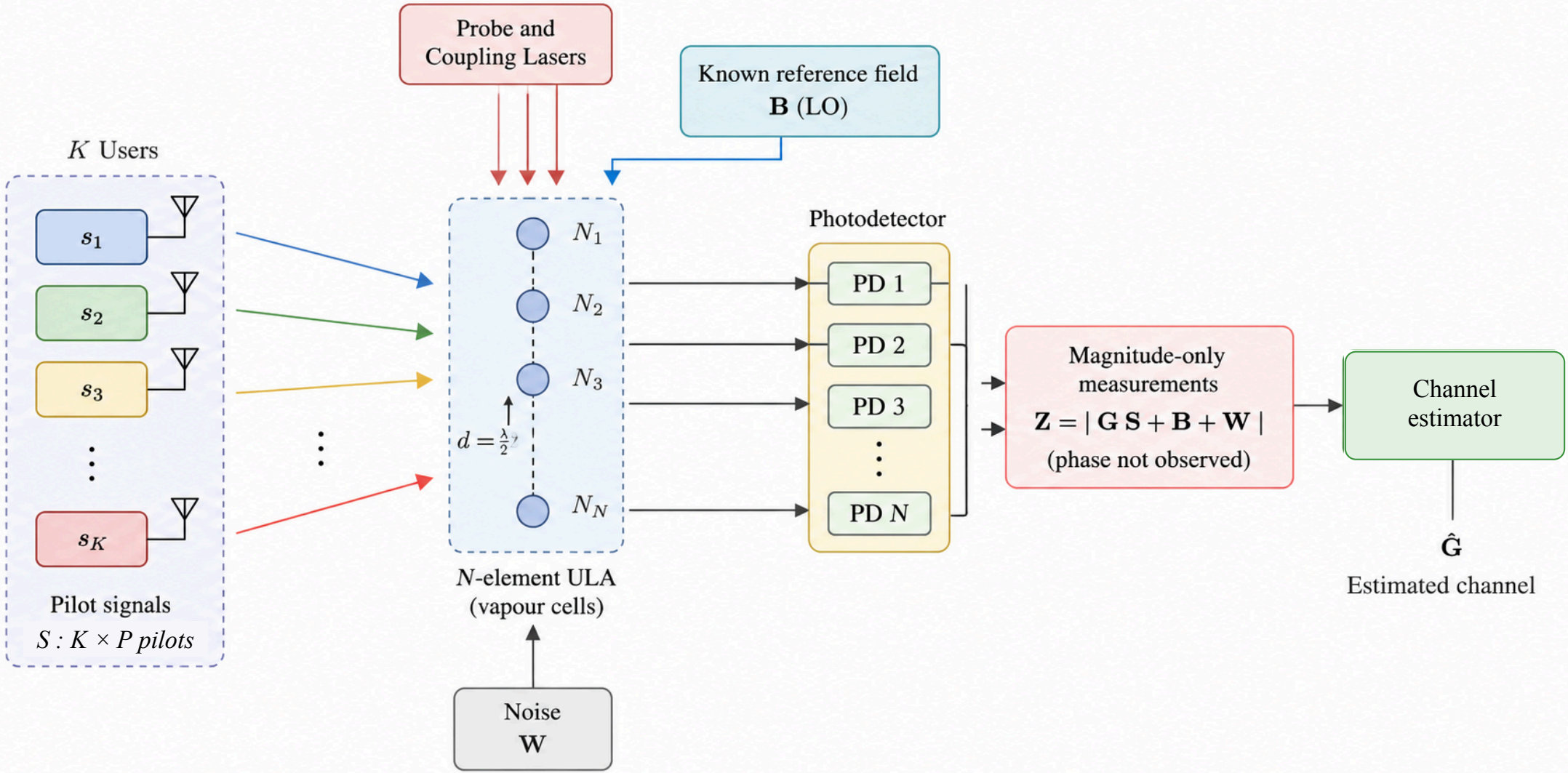

Probe and Coupling Lasers
Known reference field
B (LO)
K Users
s_1
s_2
s_3
s_K
Pilot signals
$S : K \times P$ pilots
N_1
N_2
N_3
N_N
$d = \frac{\lambda}{2}$
N-element ULA
(vapour cells)
Noise
W
Photodetector
PD 1
PD 2
PD 3
PD N
Magnitude-only measurements
$\mathbf{Z} = |\mathbf{G}\,\mathbf{S} + \mathbf{B} + \mathbf{W}|$
(phase not observed)
Channel estimator
$\hat{\mathbf{G}}$
Estimated channel